

Data and Decisions

Fall 2026

The Unemployment Rate

In-Class Team Assignment

Names:

Team:

Setup (2 minutes): one laptop. From the bCourses Class 3 page download “CPS.xlsx”: 139,931 civilian, non-institutionalized adults from the March 2019 Current Population Survey — the monthly survey of about 60,000 households behind the official U.S. unemployment rate. Use any AI that can run code. Today the AI computes outputs; your team audits the definitions, the interpretation, and the decision.

1. What is the unemployment rate?

Give your AI the dataset and this exact request: **“Compute the unemployment rate and a 95% confidence interval for it. Show exactly which Employment Status categories you counted as unemployed, which as employed, and what denominator you used.”**

- a. Getting this right requires definitional choices, not just arithmetic. Check the AI’s category mapping against the seven Employment Status categories: who did it count as unemployed? As employed? Who is excluded from the denominator entirely? Did it get it right?

- b. A colleague computes $3,579 \div 139,931 = 2.6\%$ and says “unemployment is even lower than reported!” In one or two sentences: why is that not the unemployment rate?

2. How much should you trust the second decimal?

- a. The 95% confidence interval is roughly [3.98%, 4.24%]. Write the one-sentence version for the CEO of a staffing firm, with a minimum of statistical lingo.

- b. Now ask your AI: **“The BLS published an official unemployment rate for March 2019. Does your number match? Why or why not?”** What does the gap teach you about that ± 0.13 -point interval?

3. Decide whether to launch premium service

Your staffing firm will launch a premium service this year only if average personal income among U.S. adults is at least \$41,500 — below that, the model says the price point fails. Ask your AI: “Compute average Total personal income and a 95% confidence interval.”

- a. Does the point estimate clear the bar? Does the confidence interval? What do you tell the CEO — go, no-go, or something else?

- b. Before you sign that memo, ask the AI what is actually inside that average. Is the mean the right summary for this decision? One sentence.

Before you hand this in: add your team’s AI-use statement (a few sentences — which tools you used and what you used them for).